

e-Metrology Website — Bug Report

Project: SIH26036 — e-Metrology (Digital Verification, Legal Metrology)

Site tested: <https://emetrolgy.vercel.app/>

Report date: 14 September 2026

Sources: Live playtesting by team (WhatsApp group "SIH 2k26", 13–14 Sept) + static page review

Summary

#	Bug	Module	Severity	Source
1	No input validation on business registration fields	Business Registration	Critical	Playtest
2	Contact number field allowed unlimited digits	Business Registration	Medium	Playtest
3	Weak / no password validation at registration	Authentication	High	Playtest
4	No input validation on Register Instrument fields	Instrument Registry	Critical	Playtest
5	Capacity/unit field is free text, not a controlled dropdown	Instrument Registry	Medium	Playtest
6	"Sample certificate" links on Verify page go nowhere	Public Verify	Medium	Page review
7	Inconsistent / duplicated page titles	Global	Low	Page review
8	Registration now requires live GPS with no visible manual fallback	Business Registration	Medium	Page review
9	Mobile number limit unverified end-to-end	Business Registration	Info	Page review

Bugs 1–5 were confirmed by hands-on testing (typing into the live forms and submitting) by a teammate on 13–14 Sept. Bugs 6–9 were spotted from a static review of the public pages on 14 Sept and could not be submit-tested the same way — treat them as "worth checking" rather than fully confirmed.

Section A — Confirmed bugs (live playtesting, 13–14 Sept)

Bug 1 — No input validation on business registration form

Module: Business Registration **Severity:** Critical

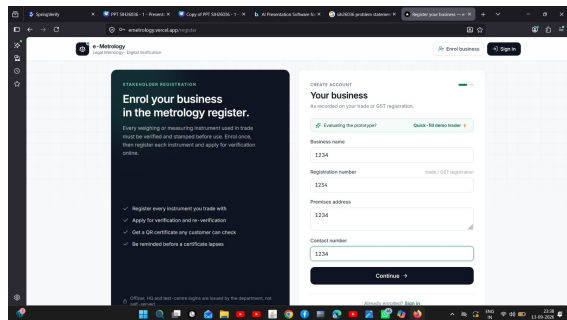
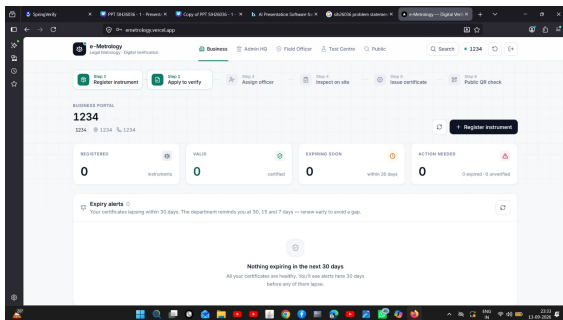
Description: All fields on the “Your business” registration step — Business name, Registration number, Premises address, and Contact number — accepted arbitrary input with no format or type validation. Testing with “1234” in every field, and separately a single letter (“m”), was accepted and successfully created a business record.

Steps to reproduce:

1. Go to emetrology.vercel.app/register.
2. Enter “1234” in Business name, Registration number, Premises address, and Contact number.
3. Click Continue.
4. Observe the business is registered and shown on the Business Portal dashboard with these values intact.

Reporter's note: “Found a bug. Name mein digits enter karne pe bhi register ho gaya. / Yeh sabh allow kar raha hai.”

Suggested fix: Add field-level validation: business name (letters/spaces, minimum length), registration number (expected GST/trade-reg format), address (minimum length), and reject purely numeric or single-character entries.



Evidence: screenshot(s) from live testing

Bug 2 — Contact number field allows unlimited digits

Module: Business Registration **Severity:** Medium

Description: The contact number input had no maximum length; it accepted an indefinite number of digits instead of being capped at 10.

Reporter's note: “contact number ma indefinite digits allowed che, if possible make it 10”

Suggested fix: Set maxlength=10 (and minlength=10) with numeric-only input and Indian mobile number format validation.

Bug 3 — No password strength validation at registration

Module: Authentication **Severity:** High

Description: The register form accepted weak, low-entropy passwords with no rule enforcement. A password of “12341234” was accepted as valid.

Reporter's note: “password rules bhi nahi hai. Maine randomly 12341234 enter kiya or yeh isko valid password ginn raha hai”

Suggested fix: Enforce a minimum password policy (e.g. 8+ characters, mix of letters/numbers/symbols) with real-time client-side feedback and server-side re-validation.

Bug 4 — No input validation on Register Instrument form

Module: Instrument Registry **Severity:** Critical

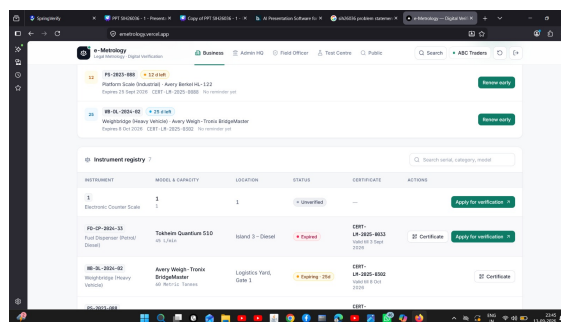
Description: Same class of issue as Bug 1, on the instrument registration form. Fields for instrument name, model, capacity, and location all accepted the value “1”, and the record was saved and listed in the Instrument Registry table with these placeholder values.

Steps to reproduce:

1. Go to Business Portal -> Register instrument.
2. Enter “1” in the instrument name, model & capacity, and location fields.
3. Submit.
4. Observe the instrument is listed in the registry with status “Unverified” and all fields showing “1”.

Reporter's note: “Yeh register instrument mein bhi yahi problem hai. See 1 1 1”

Suggested fix: Apply the same field validation approach as Bug 1: minimum length, reject placeholder/garbage values, required format checks per field.



Evidence: screenshot(s) from live testing

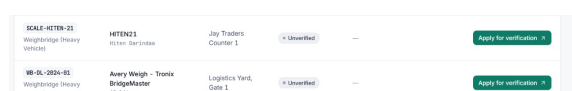
Bug 5 — Capacity/unit field is unrestricted free text

Module: Instrument Registry **Severity:** Medium

Description: The instrument capacity/unit field is a plain text box instead of a dropdown of valid units. This allowed a non-unit string (a person's name, “Hiten Darindaa”) to be saved as the unit for one instrument, and allowed “60 litres” (a volume unit) as the capacity for a Weighbridge (Heavy Vehicle), where a weight unit is expected.

Reporter's note: “Register Instrument mein hi capacity ke unit ke liye drop down de sakte hai warna koi bhi mann ganat unit likh dega... 60 litres bhi accept kar liya. To avoid spelling mistakes and random words as units, I feel drop down will be better.”

Suggested fix: Replace the free-text unit field with a dropdown scoped to the selected instrument category (e.g. Weighbridge -> kg/tonnes; Fuel Dispenser -> L/min).



Evidence: screenshot(s) from live testing

Informational notes (not bugs)

- Duplicate-email registration was tested and correctly blocked with an “account already exists” message — working as intended.
- The AI pass/fail prediction described in the PPT was confirmed present and working (“Yup it gives prediction!”).

Enhancement requests raised in the same discussion

1. Delete instrument option: No way to remove instruments after entering test/garbage data during QA. A delete icon/action is needed in the Instrument Registry.

“Ek delete instrument ka bhi icon hona chahiye. Maine itne saare random inputs daale hai unko hataane ki bhi ek facility honi chahiye.”

2. Failure-reason dropdown for onsite inspection: When a Field Officer marks an instrument as “Fail”, there is no structured way to record why. A dropdown of possible failure reasons was requested.

“Onsite inspection ke baad agar instrument ko Officer fail kar raha hai toh kyu kar raha hai uske possible reasons ka bhi dropdown hona chahiye.”

Section B — Additional findings from static page review (14 Sept)

These were spotted by inspecting the rendered public pages (Home, Sign in, Register, Verify) without logging in or submitting forms. They have not been confirmed by interactive testing — flagged here so the team can verify them directly.

Bug 6 — “Sample certificates” links on the Verify page don't go anywhere

Module: Public Verify page **Severity:** Medium

Description: On /verify, the three sample-certificate cards (Genuine, Expired, Tampered) — meant to let a visitor try verification without a printed QR sticker — all link to “#” in the rendered page instead of loading a sample certificate.

Suggested fix: Confirm whether these are meant to be live demo links; if so, wire them to real sample certificate records/routes.

Bug 7 — Inconsistent / duplicated page titles

Module: Global (browser tab titles) **Severity:** Low

Description: Browser tab titles are inconsistent across pages: /login and /register show the site name twice (e.g. “Sign in — e-Metrology · e-Metrology”), while /verify shows it once (“Verify a certificate · e-Metrology”).

Suggested fix: Standardize the title template (e.g. “[Page name] · e-Metrology”) across all routes.

Bug 8 — Registration requires live GPS with no visible manual fallback

Module: Business Registration **Severity:** Medium

Description: The registration flow now includes a required “Live location” step (“No location captured yet” until “Use my live location” is tapped). No manual address/coordinate entry is visible if geolocation is denied, unavailable, or the person isn't physically at the premises — this could block registration entirely in those cases.

Suggested fix: Verify behavior when location permission is denied or unsupported, and consider a manual fallback (e.g. pin-on-map or manual lat/long entry) with a note that officers will still verify physical presence separately.

Bug 9 — 10-digit mobile number limit unverified end-to-end

Module: Business Registration **Severity:** Info

Description: The register page currently shows a “0/10” character counter next to the mobile number field, suggesting Bug 2 (unlimited digits) may already be partially addressed. This could not be confirmed by actually submitting a longer number without interactive access.

Suggested fix: Re-test by submitting an 11+ digit number to confirm the limit is enforced, not just displayed.

Compiled from the “SIH 2k26” WhatsApp group chat (13–14 Sept 2026) and a static review of <https://emetrology.vercel.app/> on 14 Sept 2026, for the development team's reference.